UMSDTI: Unifying Multiple Molecular and Sequence Perspectives for Drug-Target Interaction Prediction—-Supplementary Materials

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A1. The initialization of atomic and bond features for molecular graphs

Our model employs node-central graph neural networks and edge-central graph neural networks, which are dedicated to extracting atomic features and chemical bond features of molecular graphs, respectively. In this process, it is essential to initialize the features of atoms and chemical bonds within the molecular graph. We utilize RDKit to swiftly extract the chemical properties associated with different atoms and chemical bonds, encoding them into features comprehensible to the model. Following previous work [1], our model's initial atom and bond features are listed in Tables A1 and A2.

Supplementary Table A1: Atom features.

Feature	Description	Size
atom type	type of atom (ex. C, N, O), by atomic number	100
bonds	number of bonds the atom is involved in	6
formal charge	integer electronic charge assigned to atom	5
chirality	unspecified, tetrahedral CW/CCW, or other	4
Hs	number of bonded hydrogen atoms	5
hybridization	sp, sp2, sp3, sp3d, or sp3d2	5
aromaticity	whether this atom is part of an aromatic system	1
atomic mass	mass of the atom, divided by 100	1

Supplementary Table A2: Bond features.

Features	Description	Size
bond type	single, double, triple, or aromatic	4
conjugated	whether the bond is conjugated	1
in ring	whether the bond is part of a ring	1
stereo	none, any, E/Z or cis/trans	6

Notably, in the features of atoms and chemical bonds, all attributes are encoded using one-hot encoding, except for atomic mass. The atomic mass is an actual number scaled to ensure the order of magnitude compatibility.

A2. Dataset Partition

A. Standard Settings

In this step, we present the detailed partition of all datasets under standard settings, including the distribution ratios of training, validation, and test sets, and how the model is trained on these datasets.

Human and C.elegans datasets are generated based on the highly credible negative DTI samples using a systematic screening framework [2]. For Human [2] and C.elegans [2] datasets, since it has not been pre-divided into training and testing sets, we follow the previous study [3] and split the dataset into training, validation, and testing sets at a ratio of 8:1:1.

The BindingDB dataset, collected from the BindingDB database [4], focuses on the interactions of small molecules. For the BindingDB [4] dataset, as the training, validation, and testing sets have already been pre-divided, we directly use the original splits without performing any additional partitioning. The statistical data of the dataset is presented in the Table A3.

Supplementary Table A3: Statistics of BindingDB dataset.

Datasets	Positive	Negative	Interactions
Train set	28237	21912	50149
Valid set	2830	2774	5604
Test set	2705	2800	5505

The GPCR dataset is created from the GLASS database [5], where both positive and negative samples have been verified through experiments. The DrugBank dataset obtained from [3] uses approved data from DrugBank release 5.0.3 (before October 2016) as the training set and newly approved data after October 2016 as an independent test set. For the GPCR [6] and DrugBank [3] dataset, the training and test sets have already been extracted and partitioned. We further re-divided the training set into a new training set and validation set at a ratio of 8:2. The statistical details of these datasets are provided in Table A4 and Table A5.

Supplementary Table A4: Statistics of GPCR dataset.

Datasets	Positive	Negative	Interactions
Train set	6569	7237	13806
Test set	752	785	1537

Our model is trained on the training set of each dataset split. The best model is selected based on the evaluation metric on the validation set. Subsequently, this best model is used for inference on the test set, and its performance is recorded as the experiment result. For each dataset, we repeat the experiments

Supplementary Table A5: Statistics of DrugBank dataset.

Datasets	Positive	Negative	Interactions
Train set	6312	6312	12624
Test set	949	3675	4624

ten times using different random seeds and report the average performance of these ten runs as the final result.

B. Cold-Start Settings

Following [7], we apply three cold-start splitting strategies on the reconstructed DrugBank dataset, which integrates the original training and test sets, as follows:

- **Cold-drug setting**: DTIs associated with 80% drugs are used to construct the training set, and the model predicts the DTIs related to the remaining 20% unseen drugs.
- **Cold-target setting**: DTIs associated with 80% targets are used to construct the training set, and the model predicts the DTIs related to the remaining 20% unseen targets.
- **Cold-drug-target setting**: DTIs composed of 80% drugs and 80% targets are used to construct the training set, and the model predicts interactions between the remaining 20% unseen drugs and 20% unseen targets.

For each training set, we randomly split 10% training samples as the validation set. The divisions of drugs and/or targets are randomly repeated ten times for each setting. The averaged results under three cold-start settings are reported.

A3. Hyperparameters

A. Hyperparameter Settings

During the training process of UMSDTI, we used the AdamW [8] optimizer and set different learning rates for different modules of the model to update the model weights. Meanwhile, we applied the warmup strategy [9] to accelerate the convergence of the model. The hyperparameters of the model are summarized in the Table A6.

Supplementary Table A6: Hyperparameters of UMSDTI

Hyperparameter	Value
Iterations of message passing	4
Number of CNN layers	3
Number of attention heads	8
Number of GCN layers in protein encoder	3
The kernel size in encoder for drug sequence	7
The kernel size in encoder for protein sequence	5
Drug hidden size	128
Protein hidden size	128
Batch size	64
Dropout	0.1
Attention heads for drug encoder module	8
Attention heads for interaction module	8
Learning rate for drug encoder	2e-4
Learning rate for target encoder	2e-4
Learning rate for interaction module	1e-4
Threshold for sparsifying contact map weights (α)	0.1

B. Hyperparameter Sensitive Analysis

To investigate the influence of key hyperparameters on model performance, we conducted a systematic analysis of the number of CNN layers, GCN layers, message passing iterations, and sparsification ratio α (Figure A1). Results reveal distinct trends in AUC and AUPR scores across varying configurations, with default settings (Table A6) consistently corresponding to optimal or well-balanced performance.

For the number of CNN layers (Figure A1 (a)), the model attains the highest AUC (0.867) and AUPR (0.864) with 3 layers, significantly outperforming other configurations. This indicates that a 3-layer CNN achieves an optimal balance between capturing local patterns in SMILES sequences and mitigating overfitting.

In the analysis of GCN layers (Figure A1 (b)), AUC (0.865–0.867) and AUPR (0.864–0.865) exhibit minimal fluctuations across 1 to 5 layers, reflecting the insensitivity of this hyperparameter to model performance. A 3-layer GCN is selected as the default, providing a balanced compromise: it expands the receptive field to capture richer molecular topological relationships while avoiding over-smoothing artifacts that may arise with deeper architectures.

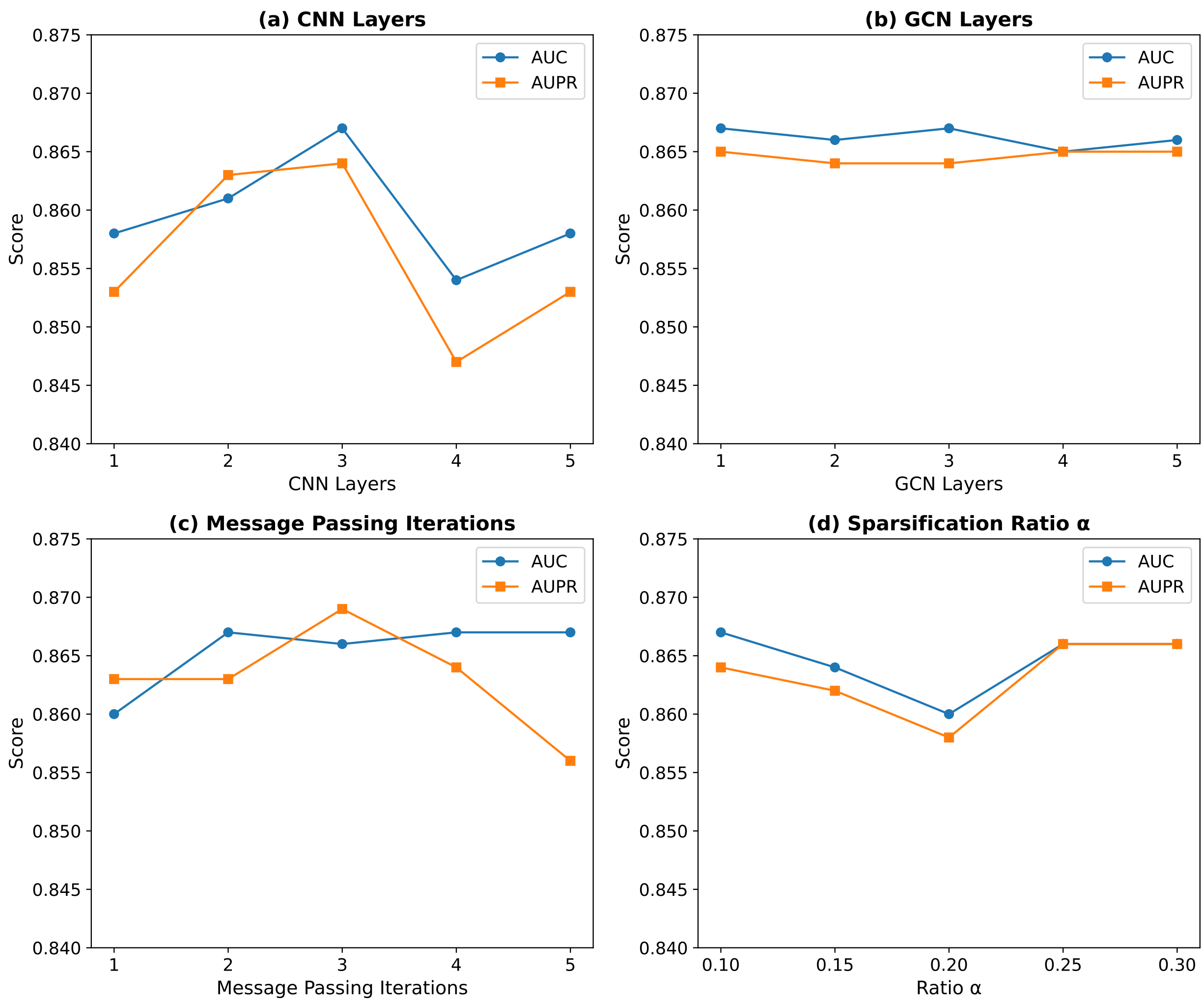
Regarding message passing iterations (Figure A1 (c)), the model sustains a high and stable AUC of 0.867 when iterations are set to 4. While AUPR peaks at 0.869 with 3 iterations, the 4-iteration configuration delivers comparable performance with enhanced stability, circumventing the performance degradation observed with 5 iterations. Thus, 4 iterations are chosen as the default to ensure robustness.

For the sparsification ratio α (Figure A1 (d)), $\alpha = 0.1$ yields the optimal AUC (0.867) and AUPR (0.864). Although $\alpha = 0.25$ and $\alpha = 0.3$ achieve similar accuracy, higher sparsity ratios incur substantial increases in computational overhead and memory consumption. Consequently, $\alpha = 0.1$ is adopted to ensure efficiency while preserving predictive performance, making it more suitable for practical training scenarios.

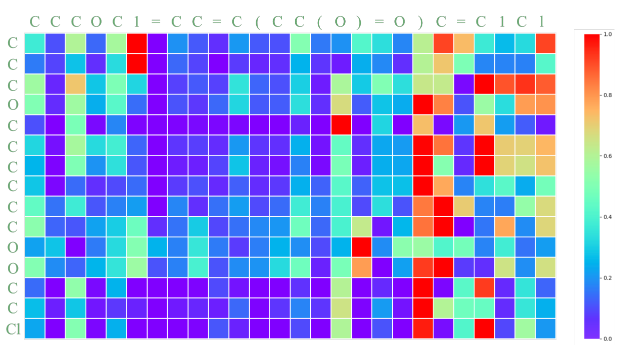
In summary, the default hyperparameter settings—3 CNN layers, 3 GCN layers, 4 message passing iterations, and sparsification ratio $\alpha = 0.1$—are empirically validated to guarantee the model's predictive efficacy and stability.

A4. Attention Interpretation in Drug Encoder

For drug feature fusion, we choose PDB:1HT8 ($C_{11}H_{13}ClO_3$) as a case study, and extract the attention matrix $\mathbf{A}^{atom} \in \mathbb{R}^{a \times s}$ learned from the cross-attention in the drug encoder (Eq.14). As shown in Figure A2, $\mathbf{A}^{atom}$ describes the association between each atom in the molecular graph (row) and each character in the SMILES string (column) during the multi-view drug feature fusion. Each atom is simultaneously associated with multiple characters in the SMILES string, allowing atoms to capture structural information from a distance. Although GNNs often overlook distant atom correlations, the cross-attention mechanism could comprehensively capture these valuable structural details.



Supplementary Figure A1: Hyperparameter sensitivity analysis of UMSDTI on GPCR dataset.



Supplementary Figure A2: Visualization of attention weights for drug feature fusion of PDB:1HT8

Supplementary Table A7: The predicted candidate targets for new drug betamethasone phosphate (DB14669)

Rank	Target name	Uniprot ID	Source
1	**Annexin A1**	**P04083**	**DrugBank**
2	**Glucocorticoid receptor**	**P04150**	**DrugBank**
3	Nitric oxide synthase	P35228	Unknown
4	Serine/threonine-protein phosphatase PP1-gamma catalytic subunit	P36873	Unknown
5	Glycogen phosphorylase, brain form	P11216	Unknown
6	Annexin A5	P08758	Unknown
7	Eukaryotic translation initiation factor 4E	P06730	Unknown
8	Caspase-8	Q14790	Unknown
9	UDP-N-acetylglucosamine 1-carboxyvinyltransferase	P33038	Unknown
10	Aldo-keto reductase family 1 member C3	P42330	Unknown

A5. Discovery of New DTIs

The top ten candidates for DB14669 and P46721 are listed in tables A7 and A8, respectively.

References

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Supplementary Table A8: The predicted candidate drugs for new target solute carrier organic anion transporter family member 1A2 (P46721)

Rank	Drug name	DrugBank ID	Source
1	**Deoxycholic acid**	**DB03619**	**DrugBank**
2	Fostamatinib	DB12010	Unknown
3	Dopamine	DB00988	Unknown
4	Ethchlorvynol	DB00189	Unknown
5	Pseudoephedrine	DB00852	Unknown
6	Flavin mononucleotide	DB03247	Unknown
7	**Cholic Acid**	**DB02659**	**DrugBank**
8	Dexanabinol	DB06444	Unknown
9	Isoflurane	DB00753	Unknown
10	**Spironolactone**	**DB00421**	**DrugBank**

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